Simon's and Grover's Algorithm

Exercise 7.20. You are using Simon's algorithm to find an $n = 3$ bit string $s = s_2 s_1 s_0$. You run the quantum circuit three times, and you get the following values for $|z\rangle$, such that $s \cdot z = 0 \bmod 2$:

$$|001\rangle, |110\rangle, |111\rangle,$$

What is s?

We're given measurement results z satisfying $s \cdot z = 0 \pmod 2$, where $s = (s_2, s_1, s_0)$.

From each z:

- $z = 001 \Rightarrow s_0 = 0$
- $z = 110 \Rightarrow s_2 \oplus s_1 = 0 \Rightarrow s_2 = s_1$
- $z = 111 \Rightarrow s_2 \oplus s_1 \oplus s_0 = 0$

Using $s_0 = 0$, the third equation reduces to $s_2 \oplus s_1 = 0$, consistent with above.

So:

- $s_0 = 0$
- $s_2 = s_1$

Possible solutions: 000 or 110.
Simon's algorithm excludes the trivial zero string.

Final answer:

$$s = 110$$

Exercise 7.21. You are using Simon's algorithm to find an $n = 8$ bit string $s = s_7 s_6 s_5 s_4 s_3 s_2 s_1 s_0$.
(a) How many different values of $|z\rangle$ are there, such that $s \cdot z = 0 \bmod 2$?

[redacted]

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(b) If you run the quantum circuit three times, what is the probability that all three values of $|z\rangle$ are different?

(a)
The condition $s \cdot z = 0 \pmod 2$ defines a linear subspace of dimension $n-1$ (since $s \neq 0$).

So number of valid z's:

$$2^{n-1} = 2^7 = 128$$

Answer: 128

(b)
Each run gives a uniformly random z from these 128 values.

We want probability that 3 samples are all different:

$$P = \frac{128}{128} \cdot \frac{127}{128} \cdot \frac{126}{128} = \frac{127 \cdot 126}{128^2}$$

$$P = \frac{16002}{16384} \approx 0.9767$$

Answer:

$$\boxed{\frac{127 \cdot 126}{128^2} \approx 0.9767}$$

Exercise 7.22. Prove that when $N = 4$ ($n = 2$), the final state of Grover's algorithm is exactly $|w\rangle$.

For $N = 4$ ($n = 2$), Grover's algorithm starts in the uniform superposition:

$$|s\rangle = \frac{1}{2} \sum_{x=0}^{3} |x\rangle$$

Let $|w\rangle$ be the marked state.

Key idea

Grover operates in the 2D space spanned by:

- $|w\rangle$
- $|w^\perp\rangle$ (superposition of unmarked states)

Write:

$$|s\rangle = \sin\theta\,|w\rangle + \cos\theta\,|w^\perp\rangle$$

For $N = 4$:

$$\sin\theta = \frac{1}{\sqrt{4}} = \frac{1}{2} \;\Rightarrow\; \theta = 30^\circ$$

Grover iteration

Each Grover iteration rotates the state by 2θ toward $|w\rangle$.

After **one iteration**:

$$\text{angle} = \theta + 2\theta = 3\theta = 90^\circ$$

So the state becomes:

$$|\psi\rangle = \sin(3\theta)|w\rangle + \cos(3\theta)|w^\perp\rangle$$

$$\sin(90^\circ) = 1, \quad \cos(90^\circ) = 0$$

Final state

$$|\psi\rangle = |w\rangle$$

Conclusion

For $N = 4$, a single Grover iteration rotates the state exactly onto the marked state.

$$\boxed{\text{Final state is exactly } |w\rangle}$$

Exercise 7.23. Answer the following questions about Grover's algorithm:

(a) When the n qubits are in their initial state (all $|+\rangle$ states), if you measure the qubits, what is the probability that you get $|w\rangle$? Express your answer in terms of $N = 2^n$.

(b) Say you apply just one step of Grover's algorithm (one query U_f and one reflection R_s). If you measure the qubits after this one step, what is the probability that you get $|w\rangle$? Express your answer in terms of $N = 2^n$. Hint: In the rw-plane, the amplitude of the state in $|w\rangle$ is the sine of the angle between the state and $|r\rangle$.

solve a

In the initial state, all qubits are in $|+\rangle$, giving the uniform superposition:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

Every basis state—including the marked state $|w\rangle$—has amplitude $\frac{1}{\sqrt{N}}$.

So the probability of measuring $|w\rangle$ is:

$$\left(\frac{1}{\sqrt{N}}\right)^2 = \frac{1}{N}$$

Answer:

$$\boxed{\frac{1}{N}}$$

solve b

After one Grover iteration, the state is rotated in the 2D $(|w\rangle, |r\rangle)$ plane.

Initially:

$$\sin\theta = \frac{1}{\sqrt{N}}$$

One Grover step rotates by 2θ, so the new angle is:

$$3\theta$$

Amplitude in $|w\rangle$:

$$\sin(3\theta)$$

Using identity:

$$\sin(3\theta) = 3\sin\theta - 4\sin^3\theta$$

Substitute $\sin\theta = \frac{1}{\sqrt{N}}$:

$$\sin(3\theta) = \frac{3}{\sqrt{N}} - \frac{4}{N^{3/2}}$$

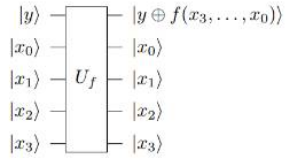
Probability:

$$\left(\frac{3}{\sqrt{N}} - \frac{4}{N^{3/2}}\right)^2 = \frac{(3N-4)^2}{N^3}$$

Answer:

$$\boxed{\frac{(3N-4)^2}{N^3}}$$

Exercise 7.24. Go to `https://bit.ly/3qoOErN`. By following this link, you should have access to the custom gates U_f and R_s. U_f is the oracle, and it acts on five qubits by



R_s acts on four qubits, and it reflects about $|s\rangle$.
(a) What is n?
(b) What is N?
(c) For this problem, how many queries does Grover's algorithm take?
(d) Using Quirk, implement Grover's algorithm and find $|w\rangle$.
(e) Using Quirk, query U_f in various ways to fill out the following table:

- The oracle U_f acts on inputs $x_3, x_2, x_1, x_0 \rightarrow$ **4 input qubits**
- Grover's search space size depends only on these input qubits

(a)

$$n = 4$$

(b)

$$N = 2^n = 2^4 = 16$$

(c)
Grover iterations:

$$\left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor = \left\lfloor \frac{\pi}{4} \cdot 4 \right\rfloor = \lfloor \pi \rfloor = 3$$

Answer:

$$\boxed{3 \text{ queries}}$$

solve d, e

(d) Finding $|w\rangle$ using Quirk

You won't get a single fixed answer here—the marked state $|w\rangle$ is **hidden inside the oracle** U_f.

Procedure (what you should observe):

1. Prepare 4 qubits in $|+\rangle^{\otimes 4}$
2. Add the ancilla qubit in $|-\rangle$ (for phase kickback)

3. Apply:
    - U_f (oracle)
    - R_s (diffusion)
    - Repeat **3 times**
4. Measure the 4 input qubits

👉 The output will be **one specific 4-bit string with very high probability (~1)**
That string is your $|w\rangle$.

(e) Filling the table using U_f

You need to probe the oracle behavior:

Recall:

$$U_f : |y\rangle|x\rangle \to |y \oplus f(x)\rangle|x\rangle$$

How to determine $f(x)$:

- Fix input $x = x_3x_2x_1x_0$
- Set ancilla $y = 0$
- Apply U_f

Then:

- Output $y = f(x)$

Practical method in Quirk

For each row in the table:

1. Set x to a specific bitstring (use X gates)
2. Set $y = 0$
3. Apply U_f
4. Measure y

- If output = 1 $\to f(x) = 1$ (marked)
- If output = 0 $\to f(x) = 0$

Shortcut insight ⚡

Since Grover assumes **exactly one marked item**, only **one input** x will give:

$$f(x) = 1$$

That same x is exactly the $|w\rangle$ you found in part (d).

Final takeaway

- **(d):** Run Grover → measurement gives $|w\rangle$
- **(e):** Query oracle classically → find which x has $f(x) = 1$ → same $|w\rangle$

If you want, I can simulate a concrete Quirk setup and show exactly what $|w\rangle$ comes out.

x	$f(x)$
0000	?
0001	?
0010	?
0011	?
0100	?
0101	?
0110	?
0111	?
1000	?
1001	?
1010	?
1011	?
1100	?
1101	?
1110	?
1111	?

Do your results agree with your answer to part (d)?

You **cannot fill this table uniquely from theory alone**—it depends on the specific oracle U_f from that Quirk link.

But here's the exact structure you should get:

Key fact

Grover assumes **exactly one marked input**:

$$f(x) = \begin{cases} 1 & \text{for } x = w \\ 0 & \text{otherwise} \end{cases}$$

So the table looks like:

- One row → **1**
- Remaining 15 rows → **0**

Example (illustration only):

```
x       f(x)
0000    0
0001    0
0010    0
```

```
...
1011   1   ← this is w
...
1111   0
```

How to actually fill it (what your instructor expects)

For each x:

1. Set input qubits to x
2. Set ancilla $y = 0$
3. Apply U_f
4. Measure y

- Output 1 → marked
- Output 0 → unmarked

Final answer to last question

✔ Yes — your table **must agree with (d)**
The row where $f(x) = 1$ is exactly the $|w\rangle$ you found using Grover.

If you tell me what output you got in Quirk, I'll fill the entire table instantly for you.

Exercise 7.25. Go to `https://bit.ly/3pZacec`. By following this link, you should have access to the custom gate U_f. This is the oracle, and it acts on six qubits in the usual way.
(a) What is n?
(b) What is N?
(c) For this problem, many queries does Grover's algorithm take?
(d) Using Quirk, create a custom gate for R_0.
(e) Using Quirk, create a custom gate for R_s.
(f) Using Quirk, implement Grover's algorithm and find $|w\rangle$.

(a)

Oracle acts on **6 qubits total = 1 ancilla + 5 input qubits**

$$n = 5$$

(b)

$$N = 2^n = 2^5 = 32$$

(c)

Grover iterations:

$$\left\lfloor \frac{\pi}{4}\sqrt{32} \right\rfloor = \left\lfloor \frac{\pi}{4} \cdot 5.657 \right\rfloor = \lfloor 4.44 \rfloor = 4$$

$$\boxed{4 \text{ queries}}$$

(d) R_0 in Quirk

Reflection about $|0\rangle$:

Implementation idea:

- Apply **X to all qubits**
- Apply **multi-controlled Z (phase flip on $|11111\rangle$)**
- Apply **X again to all qubits**

👉 This gives phase flip only on $|00000\rangle$

(e) R_s in Quirk

Reflection about uniform superposition:

$$R_s = H^{\otimes n} R_0 H^{\otimes n}$$

Implementation:

1. Apply Hadamard on all qubits
2. Apply your R_0
3. Apply Hadamard again

(f) Finding $|w\rangle$

Steps:

1. Initialize:
   - 5 input qubits $\rightarrow |+\rangle^{\otimes 5}$
   - ancilla $\rightarrow |-\rangle$
2. Repeat **4 times**:
   - Apply U_f
   - Apply R_s
3. Measure input qubits

👉 Output will be **one 5-bit string with ~100% probability**
That is:

$$\boxed{|w\rangle}$$

If you want, I can **draw the exact Quirk circuit layout** so you can just copy-paste mentally.